

LNPTM KONDUITTM COMPOUND OX10324

OX10324
REGION ASIA

DESCRIPTION

LNP* KONDUIT* OX10324 is a compound based on PPS resin. Added features of this grade include: Thermally Conductive, Electrically Conductive.

TYPICAL PROPERTY VALUES

Revision 20190717

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, brk, Type I, 5 mm/min	49	MPa	ASTM D 638
Tensile Strain, brk, Type I, 5 mm/min	0.6	%	ASTM D 638
Tensile Modulus, 5 mm/min	13500	MPa	ASTM D 638
Flexural Stress, brk, 1.3 mm/min, 50 mm span	75	MPa	ASTM D 790
Flexural Modulus, 1.3 mm/min, 50 mm span	15600	MPa	ASTM D 790
Tensile Stress, break, 5 mm/min	53	MPa	ISO 527
Tensile Strain, break, 5 mm/min	0.6	%	ISO 527
Tensile Modulus, 1 mm/min	14000	MPa	ISO 527
Flexural Stress, yield, 2 mm/min	73	MPa	ISO 178
Flexural Modulus, 2 mm/min	13500	MPa	ISO 178
IMPACT			
Izod Impact, unnotched, 23°C	70	J/m	ASTM D 4812
Izod Impact, notched, 23°C	24	J/m	ASTM D 256
Izod Impact, unnotched 80°10'4 +23°C	7	kJ/m ²	ISO 180/1U
Izod Impact, notched 80°10'4 +23°C	3	kJ/m ²	ISO 180/1A
THERMAL			
HDT, 0.45 MPa, 3.2 mm	275	°C	ASTM D 648
HDT, 1.82 MPa, 6.4 mm, unannealed	258	°C	ASTM D 648
CTE, -40°C to 40°C, flow	1.32E-05	1/°C	ASTM E 831
CTE, -40°C to 40°C, xflow	2.5E-05	1/°C	ASTM E 831
Thermal Conductivity through-plane, 60*60*3mm plaque	1.3	W/m-K	ISO 22007-2
Thermal Conductivity in-plane, 60*60*3mm plaque	18	W/m-K	ISO 22007-2
Thermal Conductivity through-plane, 10*10*3mm sample	3.5	W/m-K	ASTM E 1461-07
Thermal Conductivity in-plane, 725*0.4mm disc	15	W/m-K	ASTM E 1461-07
CTE, -30°C to 80°C, flow	1.3E-05	1/°C	ISO 11359-2
CTE, -30°C to 80°C, xflow	2.39E-05	1/°C	ISO 11359-2
CTE, 23°C to 80°C, flow	1.38E-05	1/°C	ISO 11359-2
CTE, 23°C to 80°C, xflow	2.44E-05	1/°C	ISO 11359-2
Ball Pressure Test, 165°C +/- 2°C	PASSES	-	IEC 60695-10-2
HDT/Af, 1.8 MPa Flatw 80°10'4 sp=64mm	231	°C	ISO 75/Af
Relative Temp Index, Elec ⁽¹⁾	130	°C	UL 746B
Relative Temp Index, Mech w/impact ⁽¹⁾	130	°C	UL 746B
Relative Temp Index, Mech w/o impact ⁽¹⁾	130	°C	UL 746B

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
PHYSICAL			
Mold Shrinkage, flow, 24 hrs	0.1 – 0.2	%	ISO 294
Mold Shrinkage, xflow, 24 hrs	0.1 – 0.3	%	ISO 294
Density	1.75	g/cm ³	ISO 1183
Water Absorption, 23°C/24hrs	0.01	%	ISO 62-1
Water Absorption, 23°C/24hrs	0.01	%	SABIC method
ELECTRICAL			
Surface Resistivity	>2.5E+04	Ohm	ASTM D 257
Comparative Tracking Index (UL) {PLC}	4	PLC Code	UL 746A
Hot-Wire Ignition (HWI), PLC 0	≥1.2	mm	UL 746A
FLAME CHARACTERISTICS ⁽¹⁾			
UL Yellow Card Link	E207780-100960980	-	-
UL Recognized, 94V-0 Flame Class Rating	≥1.2	mm	UL 94
Glow Wire Ignitability Temperature, 1.0 mm ⁽²⁾	850	°C	IEC 60695-2-13
UV-light, water exposure/immersion	F1	-	UL 746C
INJECTION MOLDING			
Drying Temperature	120 – 150	°C	
Drying Time	4	hrs	
Melt Temperature	320 – 350	°C	
Front - Zone 3 Temperature	315 – 345	°C	
Middle - Zone 2 Temperature	315 – 345	°C	
Rear - Zone 1 Temperature	315 – 345	°C	
Mold Temperature	110 – 150	°C	
Back Pressure	0.3 – 0.7	MPa	
Screw Speed	60 – 100	rpm	
Shot to Cylinder Size	50 – 75	%	

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

(2) Value shown here is based on internal measurement.

DISCLAIMER

Any sale by SABIC, its subsidiaries and affiliates (each a “seller”), is made exclusively under seller’s standard conditions of sale (available upon request) unless agreed otherwise in writing and signed on behalf of the seller. While the information contained herein is given in good faith, SELLER MAKES NO WARRANTY, EXPRESS OR IMPLIED, INCLUDING MERCHANTABILITY AND NONINFRINGEMENT OF INTELLECTUAL PROPERTY, NOR ASSUMES ANY LIABILITY, DIRECT OR INDIRECT, WITH RESPECT TO THE PERFORMANCE, SUITABILITY OR FITNESS FOR INTENDED USE OR PURPOSE OF THESE PRODUCTS IN ANY APPLICATION. Each customer must determine the suitability of seller materials for the customer’s particular use through appropriate testing and analysis. No statement by seller concerning a possible use of any product, service or design is intended, or should be construed, to grant any license under any patent or other intellectual property right.